

Data Science

Topic: Robotics AI Intelligence: Predicting Robot Task Performance Using Artificial Intelligence

1. Data Collection

Objective

Collect data related to robot operations and intelligent decision-making to build an AI model capable of predicting robot task success and performance.

Data Sources

- Public robotics datasets from:
 - Kaggle
 - UCI Machine Learning Repository
 - Open Robotics
- Robot sensor logs
- Simulated robot environments
- IoT-enabled robotic systems

Collected Features

Feature	Description
Sensor Accuracy	Accuracy of robot sensors
Battery Level	Remaining battery percentage
Processing Speed	CPU performance
Task Complexity	Difficulty level of assigned task
Obstacle Distance	Distance to nearby obstacles
Environmental Conditions	Indoor/Outdoor conditions
Task Completion Time	Time taken to complete task
Task Success	Target variable (Success/Failure)

Data Format

- CSV Files
- JSON Logs
- Sensor Stream Data

2. Data Cleaning

Objective

Prepare raw robotics data for analysis and machine learning.

Steps Performed

Missing Value Handling

- Removed incomplete records
- Filled missing numerical values using mean/median

Duplicate Removal

- Eliminated duplicate robot activity records

Outlier Detection

- Identified abnormal sensor readings using:
 - Boxplots
 - Interquartile Range (IQR)

Data Transformation

- Converted categorical variables into numerical values
- Standardized numerical features

Example

Before Cleaning:

Battery Sensor Accuracy

80	95
Null	92

After Cleaning:

Battery Sensor Accuracy

80 95

76 92

3. Data Analysis

Objective

Understand patterns and relationships within robotics data.

Exploratory Data Analysis (EDA)

Descriptive Statistics

- Mean
- Median
- Standard Deviation
- Variance

Visualizations

- Histograms
- Boxplots
- Scatter Plots
- Correlation Heatmaps

Key Findings

1. Higher battery levels improve task success rates.
2. Sensor accuracy strongly affects robot performance.
3. Increased task complexity reduces success probability.
4. Faster processing speeds lead to quicker task completion.

Correlation Analysis

Feature	Correlation with Task Success
Battery Level	+0.75
Sensor Accuracy	+0.82
Processing Speed	+0.68
Task Complexity	-0.60

4. Model Building

Objective

Develop a machine learning model to predict robot task success.

Machine Learning Algorithms

Random Forest Classifier

Selected because:

- High accuracy
- Handles nonlinear relationships
- Robust against overfitting

Data Split

Dataset	Percentage
---------	------------

Training Data 80%

Testing Data 20%

Model Workflow

1. Import dataset
2. Select features
3. Split data
4. Train model
5. Generate predictions

Features Used

- Battery Level
- Sensor Accuracy
- Processing Speed
- Obstacle Distance
- Task Complexity

Target Variable

- Task Success (Yes/No)
-

5. Model Evaluation

Objective

Measure model performance and reliability.

Evaluation Metrics

Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

$$Precision = \frac{TP}{TP + FP}$$

Recall

$$Recall = \frac{TP}{TP + FN}$$

F1 Score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Sample Results

Metric	Value
--------	-------

Accuracy	94%
----------	-----

Precision	92%
-----------	-----

Recall	93%
--------	-----

F1 Score	92.5%
----------	-------

Confusion Matrix

	Predicted Success	Predicted Failure
Actual Success	180	12
Actual Failure	10	98

Conclusion

The Random Forest model achieved high accuracy and demonstrated strong predictive capability for robotic task performance.

6. Deployment

Objective

Deploy the AI model for real-world robotic applications.

Deployment Architecture

Robot Sensors → Data Processing → AI Model → Decision Engine → Robot Action

Deployment Tools

- Python
- Flask/FastAPI
- Docker
- Cloud Services

Deployment Steps

1. Save trained model using Pickle (.pkl)
2. Create API using Flask/FastAPI
3. Connect robot sensor inputs
4. Send data to AI model
5. Receive prediction
6. Execute robotic decision

Example Output

Input:

- Battery Level = 85%
- Sensor Accuracy = 96%
- Task Complexity = Medium

Prediction:

- Task Success Probability = 97%

Benefits

- Real-time decision making
- Improved robotic efficiency
- Reduced operational failures
- Intelligent autonomous behaviour

Conclusion

This project demonstrates how Data Science and Artificial Intelligence can enhance robotic intelligence by analyzing sensor data, predicting task outcomes, and enabling autonomous decision-making. The developed machine learning model successfully predicts robotic task success and can be deployed in industrial robots, autonomous vehicles, healthcare robots, and smart manufacturing systems.